LECTURE NOTES

Lecture 1.2: Modeling, Laplace Transforms, and Transfer Functions

Week 1 – A Simple DC Motor Model

Class Purpose

This lecture shows how a simple differential equation becomes a transfer function. The goal is not to complete a full Laplace-transform unit. The goal is to understand the model used later for DC motor speed control.

Main question for today: How can a simple mathematical model describe the speed response of a DC motor?

75-Minute Class Plan

Time	Topic or activity
0–10 min	Why engineers model systems
10–25 min	First-order differential equation model
25–40 min	Laplace-transform review and derivation
40–52 min	Transfer functions and parameter meaning
52–60 min	Pole of a first-order system
60–70 min	Worked example
70–75 min	Python connection and summary

Learning Objectives

By the end of this lecture, students should be able to:

1. explain why engineers use mathematical models;
2. interpret a simple first-order differential equation;
3. apply a basic Laplace-transform rule under zero initial conditions;
4. derive a first-order transfer function;
5. identify gain K, time constant τ, and pole location;
6. connect a transfer function to a Python simulation.

1. Why We Model Systems

The real motor system is

$$\boxed{\text{PWM}} \longrightarrow \boxed{\text{Motor}} \longrightarrow \boxed{\text{RPM}}.$$

A mathematical model represents this behavior in a simplified form:

$$U(s) \longrightarrow \boxed{G(s)} \longrightarrow Y(s).$$

Models help us:

- predict motor response;
- simulate the system before changing hardware;
- compare possible controller settings;
- understand why the motor responds quickly or slowly;
- design feedback control more safely.

> A useful model does not need to capture every detail of the physical motor. It only needs to capture the behavior needed for the engineering question.

2. A Simple First-Order Dynamic Model

Many systems do not change instantly. A motor has inertia, so its speed changes gradually.
A simple first-order model is

$$\boxed{\tau \frac{dy(t)}{dt} + y(t) = Ku(t)}$$

where

- $y(t)$ is the motor speed;
- $u(t)$ is the motor command;
- K is the system gain;
- τ is the time constant.

This model says that the motor speed depends on both the current input and how quickly the speed is changing.

3. Physical Meaning of the Model

Consider

$$\tau \frac{dy(t)}{dt} + y(t) = Ku(t).$$

The term $y(t)$ represents the present motor speed.

The term $\frac{dy(t)}{dt}$ represents the rate of change of speed.

The parameter K affects the final speed reached for a given input.

The parameter τ affects how quickly the motor approaches that final speed.

If τ is large, the motor response is slower. If τ is small, the motor response is faster.

4. Very Short Laplace Review

We will use only a few basic Laplace-transform ideas.

$$\mathcal{L}\{y(t)\} = Y(s)$$

$$\mathcal{L}\{u(t)\} = U(s)$$

Under zero initial conditions,

$$\mathcal{L}\left\{\frac{dy(t)}{dt}\right\} = sY(s).$$

In this course, the Laplace transform is a tool for building transfer functions. We will not focus on complicated transform tables.

5. Deriving the Transfer Function

Start with the first-order model:

$$\tau \frac{dy(t)}{dt} + y(t) = Ku(t).$$

Apply the Laplace transform under zero initial conditions:

$$\tau sY(s) + Y(s) = KU(s).$$

Factor out $Y(s)$:

$$Y(s)(\tau s + 1) = KU(s).$$

Divide by $U(s)(\tau s + 1)$:

$$\frac{Y(s)}{U(s)} = \frac{K}{\tau s + 1}.$$

Therefore, the transfer function is

$$\boxed{G(s) = \frac{Y(s)}{U(s)} = \frac{K}{\tau s + 1}.}$$

6. Transfer Function Definition

A transfer function is the output divided by the input in the Laplace domain:

$$\boxed{G(s) = \frac{Y(s)}{U(s)}.}$$

For the DC motor speed-control problem:

$$G(s) = \frac{\text{motor speed}}{\text{motor command}}.$$

The first-order motor model used in this course is

$$\boxed{G(s) = \frac{K}{\tau s + 1}.}$$

7. Meaning of Gain and Time Constant

Parameter	Meaning
K	Steady-state gain. It describes how much output the system eventually produces for a given input.
τ	Time constant. It describes how quickly the system responds.

A larger K means more RPM for the same command.

A smaller τ means the motor responds faster.

8. Pole of a First-Order System

The pole is found from the denominator of the transfer function.

For

$$G(s) = \frac{K}{\tau s + 1},$$

the denominator is

$$\tau s + 1.$$

Set the denominator equal to zero:

$$\tau s + 1 = 0.$$

Solve for s:

$$\boxed{s = -\frac{1}{\tau}}.$$

Example

If

$$\tau = 0.5 \text{ s},$$

then

$$s = -\frac{1}{0.5} = -2.$$

For a first-order system, the pole location is related to how quickly the system responds. Detailed stability analysis will come later.

9. Worked Example

Given the dynamic equation

$$2\frac{dy(t)}{dt} + y(t) = 4u(t),$$

find the transfer function.

Apply the Laplace transform:

$$2sY(s) + Y(s) = 4U(s).$$

Factor $Y(s)$:

$$Y(s)(2s + 1) = 4U(s).$$

Divide by $U(s)(2s + 1)$:

$$G(s) = \frac{Y(s)}{U(s)} = \boxed{\frac{4}{2s + 1}}.$$

Identify the parameters:

$$K = 4, \qquad \tau = 2.$$

The pole is

$$s = -\frac{1}{\tau} = -\frac{1}{2}.$$

10. Another Practice Problem

For the model

$$0.25\frac{dy(t)}{dt} + y(t) = 1200u(t),$$

answer the following:

1. What is K?
2. What is τ?
3. What is the transfer function $G(s) = Y(s)/U(s)$?
4. What is the pole location?
5. Would this response be faster or slower than a system with $\tau = 2$?

11. Python Connection

Python can represent the transfer function directly.
For

$$G(s) = \frac{K}{\tau s + 1},$$

one possible Python script is:

```
import numpy as np
import matplotlib.pyplot as plt

K = 4
tau = 2

t = np.linspace(0, 12, 400)
y = K * (1 - np.exp(-t / tau))

print("pole =", -1 / tau)

plt.plot(t, y, linewidth=2)
plt.grid(True)
plt.xlabel("Time (s)")
plt.ylabel("Output")
plt.title("First-Order Step Response")
plt.show()
```

The mathematics gives us the transfer function. Python helps us visualize the response.

Summary

- A model is a simplified representation of a physical system.
- A useful first-order motor model is

$$\tau \frac{dy(t)}{dt} + y(t) = Ku(t).$$

- Under zero initial conditions, the transfer function is

$$G(s) = \frac{K}{\tau s + 1}.$$

- K describes final response magnitude.
- τ describes response speed.
- The pole of the first-order system is $s = -1/\tau$.